
RENTASALUD: A WEB-BASED INTERACTIVE ATLAS OF SOCIAL INEQUALITIES AND LIFE EXPECTANCY IN ANDALUSIA (SOUTHERN SPAIN)

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ABSTRACT

Background Interactive tools linking income and life expectancy at fine geographical scales are scarce for southern Europe. **Methods** We linked INE census-section income data (48,262 section-years, 2015–2022) with BDLPA cohort life tables (~637,000 individuals, follow-up 2011–2023) for Andalusia, Ceuta, and Melilla. Correlation and clustering assessed the income–longevity association. **Results** Life expectancy was 79.5 (men) and 84.4 years (women). Eliminating tumours would yield the largest gain for men (3.93 years), while heart disease accounted for the largest gain among women (2.89 years). The income–life expectancy correlation across ten provinces was weak ($r = 0.28$, 95%CI 0.42 to 0.77, $P = 0.43$; Spearman $\rho = 0.24$). Provincial EV ranged from 79.4 (Almería) to 82.7 years (Córdoba), a gradient of 3.3 years. An R Shiny application integrates all findings in six interactive tabs. **Conclusion** RENTASALUD provides an open tool for exploring health inequalities. The weak provincial income–EV correlation is compatible with the much steeper census-tract gradients reported nationally. Aggregation at the province level attenuates the association; income outliers (Ceuta, Melilla) further weaken it. **Keywords:** health inequalities; life expectancy; spatial epidemiology; interactive atlas; Shiny; Andalusia * **Equal contribution.** GRG and PMG contributed equally as co-first authors.

1 Introduction

People with higher incomes live longer, on average, across most European countries (Marmot 2010; Mackenbach et al. 2015). In Southern Europe the gap between rich and poor is narrower than in the north or east, a pattern that researchers have connected to the Mediterranean diet, strong family and community ties, and universal healthcare (Mackenbach et al. 2018; Regidor et al. 2012). Even so, educational disparities in Spain still drive roughly one in five deaths each year — some 80,000 people — mostly from heart disease, lung cancer, respiratory illness, and accidents (Regidor et al. 2014).

When the analysis moves down to small areas like neighbourhoods or census tracts, the picture sharpens. Redondo-Sánchez and colleagues (Redondo-Sánchez et al. 2022) found that Spanish women and men in the most deprived neighbourhoods lived 3.2 and 3.8 fewer years, respectively, than those in the wealthiest areas. The gap was wider for men and varied across the country; the north and provincial capitals did better.

Andalusia is Spain’s largest region, home to eight and a half million people, and one of its poorest. Plenty of maps exist, but nearly all of them are either too coarse (whole provinces) or too static to be useful to a town councillor, a public health officer, or a curious resident. The aim of our project, RENTASALUD, was to fix that: to build a single tool where income, demographics, and life expectancy can be explored together at the neighbourhood scale, with a year slider and an AI assistant that answers questions in plain language.

RENTASALUD is an open source web atlas. It draws on census-section data from Spain’s National Statistics Institute (INE, 2015–2022) and life expectancy estimates from the Longitudinal Database of the Andalusian Population (BDLPA), which tracks roughly 637,000 people from the 2011 census through to the end of 2023. The analysis covers the eight Andalusian provinces plus Ceuta and Melilla. We set out to estimate life expectancy by province and sex, to quantify how much each cause of death shortens life, to examine whether provincial income predicts longevity, and to put it all in an interactive tool that requires no statistical training to use.

2 Methods

2.1 Data sources

The income and demographic data come from the INE *Atlas de Distribución de Renta de los Hogares* (Instituto Nacional de Estadística 2024), which covers the eight Andalusian provinces and the autonomous cities of Ceuta and Melilla for the period 2015–2022. It contains over 48,000 census section–year observations: median income, age structure, household size and composition, and total population.

The mortality data come from the BDLPA (Instituto de Estadística y Cartografía de Andalucía 2024), a random 10% sample of everyone who lived in Andalusia at the 2011 census, followed through December 2023. For each person the database records whether they were still alive, had moved away, or had died. When a death occurred, the cause is coded into one of ten broad groups: heart and circulatory disease, malignant tumours, diabetes and endocrine conditions, infectious disease, respiratory disease, digestive disease, nervous system disorders, external causes (accidents and suicide), alcohol-related causes, and all other causes.

2.2 Life expectancy estimation

We used standard abridged life table methods (Chiang 1968) with 5-year age bands (0–4, 5–9, . . . , 85–89) and an open final band for ages 90 and over. Cause-deleted life tables (Beltrán-Sánchez et al. 2008) were computed to estimate the gain in life expectancy at birth that would result from removing each cause, one at a time, using Chiang’s proportional-risk adjustment. Life expectancy was calculated at the province level (8 provinces + 2 autonomous cities $\times$ 2 sexes = 20 life tables). Census-tract estimation was not possible because the BDLPA identification numbers are sequential codes with no geographic information beyond the two-digit province code.

2.3 Income and life expectancy

We examined the province-level association between median income and mean life expectancy using Pearson and Spearman correlations, plus hierarchical clustering to explore groupings. With only ten data points the analysis is exploratory and has limited statistical power. As a sensitivity check we repeated the correlation after removing Ceuta and Melilla — autonomous cities whose economies depend heavily on cross-border trade and public-sector employment, making them structurally different from the Andalusian provinces.

2.4 Ethics

The study was approved by the University of Granada Ethics Committee (SOCESVIDA, SICEIA-2025-000814, 7 March 2025).

3 Results

3.1 Life expectancy in Andalusia

A newborn in Andalusia can expect to live about 79.5 years if male and 84.4 if female, a gap of just under 5 years. The difference reflects higher death rates from heart disease, lung cancer, and accidents among men. As age increases, remaining life expectancy drops in the expected pattern: at 65, a man has about 19.9 years left and a woman about 23.5, while at 85 the remaining years are 7.3 and 8.4.

Life expectancy differs across the ten territories (Table 1). Almería sits at the bottom for both sexes (77.8 years for men, 81.1 for women); Córdoba is at the top (80.1 and 85.3). The gap between lowest and highest is roughly 2.3 years for men and 4.2 for women. Ceuta and Melilla fall in the middle of the range: Ceuta records 79.8 years for men and 84.7 for women, Melilla 79.8 and 85.1. Their incomes (16,899 and 15,443) are well above the Andalusian average of roughly 13,300, yet their life expectancy is not proportionally higher, a mismatch that pulls the overall income–EV correlation downwards.

Table 1: Life expectancy at birth by province and sex (ordered by male life expectancy ascending). Ceuta and Melilla are autonomous cities.

Province	Men (years)	Women (years)
Almería	77.8	81.1
Jaén	79.7	84.9
Málaga	79.6	84.8
Ceuta	79.8	84.7
Melilla	79.8	85.1
Granada	79.8	85.1
Huelva	79.9	84.5
Cádiz	80.0	85.0
Sevilla	80.0	85.2
Córdoba	80.1	85.3

3.2 Cause elimination

Removing cancer from the mortality profile would add about 3.9 years to male life expectancy at birth and 2.8 to female (Table 2). For women, eliminating heart and circulatory disease would have a slightly larger impact at 2.9 years. Taken together, tumours and cardiovascular disease account for 7.1 years of potential gain for men and 5.7 for women.

Table 2: Years gained at birth if each cause were eliminated.

Cause	Men (years)	Women (years)
Malignant tumours	3.93	2.83
Heart and circulatory diseases	3.15	2.89
Respiratory diseases	1.13	0.69
External causes (accidents, suicide)	0.67	0.33
Digestive diseases	0.57	0.48
Nervous system diseases	0.50	0.57
Infectious diseases	0.41	0.35
Diabetes and endocrine diseases	0.28	0.32
Alcohol-related causes	0.07	0.02
Other causes	1.13	1.19

Men lose 1.1 more years to cancer than women do, consistent with the historically heavier smoking burden among Spanish men. Alcohol-related causes barely register at the population level, although their toll on affected individuals and families can be severe.

3.3 Income and life expectancy

At the province level, the linear association between median income and mean life expectancy was weak and not statistically distinguishable from zero ($r = 0.28$, 95% CI: 0.42 to 0.77, $P = 0.43$; Spearman $= 0.24$, $P = 0.51$). Ceuta (16,899 median income, 82.2 years mean EV) and Melilla (15,443, 82.4 years) stand out: their incomes substantially exceed the Andalusian range of 12,699–14,338, yet their life expectancy sits squarely in the middle, pulling the regression line toward flat. Removing these two cities lifts the correlation to $r = 0.52$ ($P = 0.19$). The sign is positive, but with ten data points the confidence interval is wide enough to be compatible with anything from a moderate negative association to a strong positive one. The 3.3-year spread in mean provincial life expectancy (79.4 in Almería to 82.7 in Córdoba) does indicate a real territorial gradient, even if its statistical expression is noisy.

The weak aggregate association does not imply that income is irrelevant to health. At the neighbourhood level, Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022) found a steep, precise gradient: 3.8 years for men and 3.2 for women between the extremes of deprivation. The gap between a provincial r of 0.28 and a neighbourhood-level gulf of 3–4 years is exactly what one would expect when summarising thousands of census tracts into ten administrative units, particularly when two of those units operate under different economic rules. Neighbourhood-scale data are needed to see the gradient; tools like RENTASALUD are designed to show where the sharpest inequalities lie.

3.3.1 Sensitivity to Ceuta and Melilla

Ceuta and Melilla are autonomous cities whose economies are shaped by cross-border trade, a large public sector, and special tax arrangements, rather than by the agricultural and service activities that dominate Andalusia. To gauge their influence, we repeated the correlation analysis on the eight Andalusian provinces alone (Table 3).

Table 3: Income–life expectancy correlation with and without Ceuta and Melilla.

	10 territories	8 provinces (Andalusia only)
Pearson r (95% CI)	0.28 (0.42, 0.77)	0.52 (0.30, 0.90)
P-value	0.43	0.19
Spearman	0.24 ($P = 0.51$)	0.48 ($P = 0.24$)
R^2	0.08	0.27
EV spread (mean)	3.3 years	3.4 years

Excluding the two cities more than triples the explained variance (R^2 from 0.08 to 0.27) and doubles the correlation coefficient. The result remains non-significant at conventional thresholds, but the shift is large enough to suggest that the income–longevity link operates differently in territories with disparate economic structures. Aggregate ecological analyses are sensitive to the inclusion of heterogeneous units, something worth bearing in mind when only a handful of spatial units are available.

3.4 The RENTASALUD application

All the results described above are accessible through an interactive R Shiny application with six tabs:

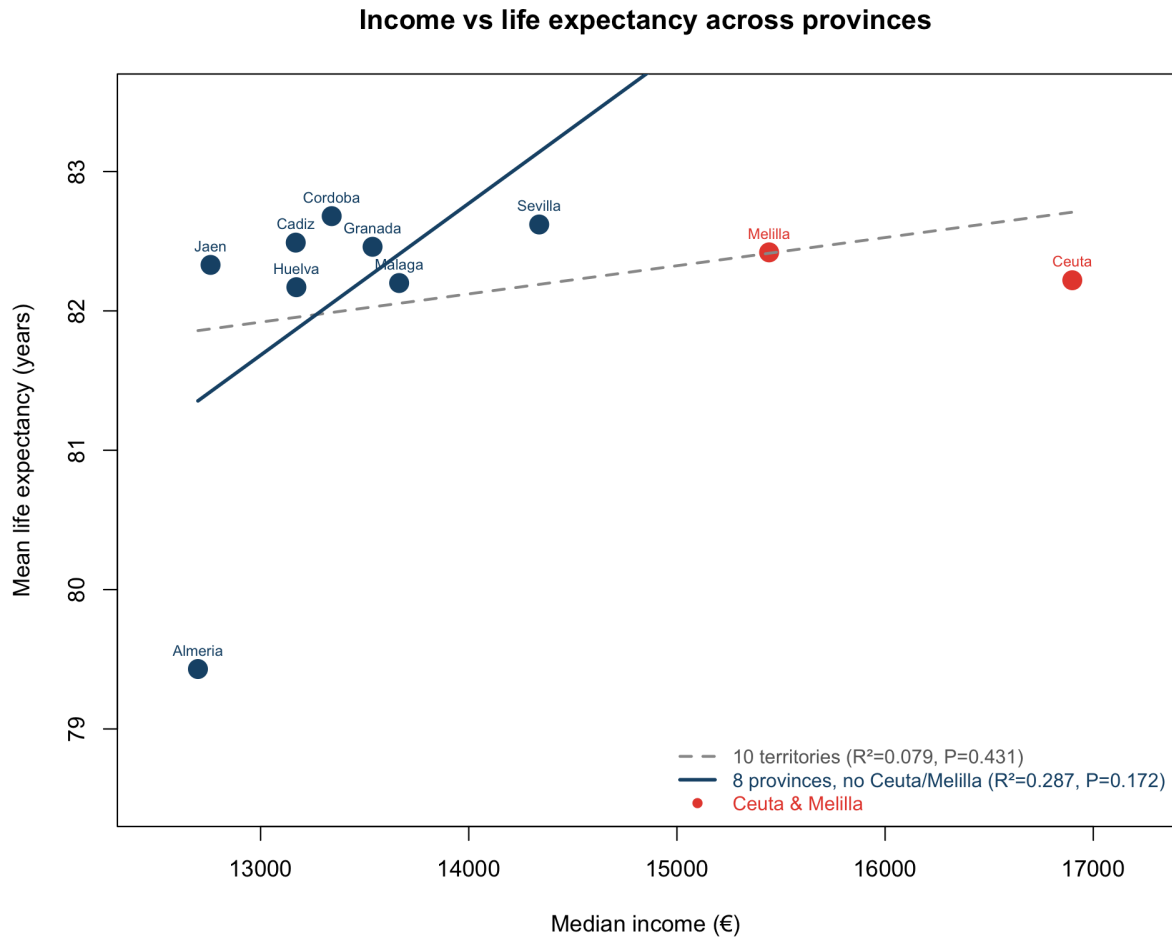


Figure 1: Median income and life expectancy across provinces. Blue points: Andalusian provinces. Red points: Ceuta and Melilla (income outliers). Solid blue line: linear fit excluding Ceuta and Melilla ($R^2 = 0.29$, $P = 0.17$). Dashed grey line: fit including all ten territories ($R^2 = 0.08$, $P = 0.43$).

- **Explore** — choropleth maps of income, demographics, and life expectancy at the census-section level, with a year slider (2015–2022).
- **AI Assistant** — natural-language queries in Spanish, answered by a large language model running locally on the user’s machine or via a cloud API.
- **Time Series** — yearly trends for any indicator, filtered by province.
- **Relationships** — scatter plots and bar charts exploring income and life expectancy, with and without Ceuta and Melilla.
- **Mortality Causes** — interactive charts and tables showing gains from eliminating each cause.
- **Methodology** — indicator definitions, data sources, and citation details.

The full analysis code is available at github.com/migariene/RENTASALUD. The live application runs at watzile.shinyapps.io/RENTA.

4 Discussion

4.1 What we found

Life expectancy in Andalusia, Ceuta, and Melilla sits below the Spanish average, which fits with the region’s lower income levels. Cancer and heart disease are the main drivers of premature death. The larger cancer burden in men (an extra 1.1 years of potential gain) mirrors Spain’s historical smoking patterns, a gap that is closing in younger generations.

4.2 Why the provincial income–EV link is weak

The province-level correlation ($r = 0.28$, $P = 0.43$) is weak and imprecise. Two things flatten it. The first is scale: thousands of census tracts aggregated into ten units, which is bound to shrink variance and lose power. The second is Ceuta and Melilla: cities with high median incomes that do not translate into proportionally longer lives. When they are removed, the correlation among the eight Andalusian provinces rises to $r = 0.52$, enough to be suggestive but still not conclusive.

A 3.3-year spread in provincial life expectancy (79.4 to 82.7) is real and, at this coarse resolution, runs in the expected direction. The steeper gradients that Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022) found at the neighbourhood level suggest that within-province differences probably exceed what we can detect here. The Southern European pattern of compressed health inequalities (Mackenbach et al. 2018; Regidor et al. 2012) does not mean inequality is absent: it means the signal operates at a finer scale and may be dampened by diet, social networks, and healthcare access. Policy efforts that target the most disadvantaged neighbourhoods, rather than entire provinces, are more likely to make a measurable difference.

4.3 Strengths and limitations

The study links individual-level follow-up on nearly 637,000 people over 12 years to neighbourhood socioeconomic data. Everything is open source and runs through an interactive platform with a built-in natural-language interface.

The most important limitation is geographic: life expectancy could only be estimated at province level because the BDLPA person identifiers carry no sub-provincial geographic code. Linking the BDLPA to the municipal register (Padrón Continuo) would unlock small-area estimation and is, in our view, the most worthwhile next step. Other limitations include the assumption of constant mortality rates, small cell sizes at extreme ages when stratified by province and sex, a partial mismatch between the income window (2015–2022) and the mortality follow-up (2011–2023), and the fact that cause-deleted life tables treat causes as independent when in practice they are not.

5 What is already known on this topic

- In Spain, residents of the most deprived neighbourhoods live 3 to 4 years less than those in the wealthiest.
- Southern Europe has smaller health inequalities than Northern or Eastern Europe.
- Fine-scale interactive tools that let users explore income alongside life expectancy are rare.
- Autonomous cities with border-trade economies (Ceuta, Melilla) may sit outside typical income–health gradients.

6 What this study adds

- RENTASALUD is the first open source interactive atlas of neighbourhood-level income and life expectancy for Andalusia, Ceuta, and Melilla.
- At province level, the income–life expectancy correlation is weak ($r = 0.28$, $P = 0.43$) and doubles when Ceuta and Melilla are excluded ($r = 0.52$).
- Mean provincial life expectancy spans 3.3 years (79.4 in Almería to 82.7 in Córdoba), and the gap is wider for women.
- Cancer and cardiovascular disease together account for 7.1 years of potential life expectancy gain in men and 5.7 in women.
- A built-in AI assistant lets users query the data in everyday language with no technical training.

7 Acknowledgements

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8 Data Availability

Analysis code: github.com/migariene/RENTASALUD. Shiny application: watzile.shinyapps.io/RENTA. BDLPA microdata available from IECA on request.

References

Beltrán-Sánchez, Hiram, Samuel H. Preston, and Vladimir Canudas-Romo. 2008. “An Integrated Approach to Cause-of-Death Analysis: Cause-Deleted Life Tables and Decompositions of Life Expectancy.” *Demographic Research* 19: 1565–602. <https://doi.org/10.4054/DemRes.2008.19.35>.

- Chiang, Chin Long. 1968. *Introduction to Stochastic Processes in Biostatistics*. Wiley.
- Instituto de Estadística y Cartografía de Andalucía. 2024. *Estadísticas Longitudinales de Supervivencia y Longevidad En Andalucía: Ficheros de Microdatos, Cohorte 2011*. <https://www.juntadeandalucia.es/institutodeestadisticaycartografia/bdlpa/>.
- Instituto Nacional de Estadística. 2024. *Atlas de Distribución de Renta de Los Hogares: Metodología*. https://www.ine.es/dyngs/INEbase/es/operacion.htm?c=Estadistica_C&cid=1254736177088&menu=metodologia.
- Mackenbach, Johan P., Ivana Kulhánová, Gwenn Menvielle, et al. 2015. “Trends in Inequalities in Premature Mortality: A Study of 3.2 Million Deaths in 13 European Countries.” *Journal of Epidemiology and Community Health* 69 (3): 207–17. <https://doi.org/10.1136/jech-2014-204319>.
- Mackenbach, Johan P., José R. Valverde, Barbara Artnik, et al. 2018. “Trends in Health Inequalities in 27 European Countries.” *Proceedings of the National Academy of Sciences* 115 (25): 6440–45. <https://doi.org/10.1073/pnas.1800028115>.
- Marmot, Michael. 2010. *Fair Society, Healthy Lives: The Marmot Review*. University College London.
- Redondo-Sánchez, Daniel, Mara-José Sánchez, Pablo Fernández-Navarro, Bernard Rachet, and Miguel Angel Luque-Fernandez. 2022. “Association of Socioeconomic Deprivation with Life Expectancy and All-Cause Mortality in Spain, 2011-2013.” *Scientific Reports* 12 (1): 15554. <https://doi.org/10.1038/s41598-022-19859-1>.
- Regidor, Enrique, Anton E. Kunst, Fernando Rodríguez-Artalejo, and Johan P. Mackenbach. 2012. “Small Socio-Economic Differences in Mortality in Spanish Older People.” *European Journal of Public Health* 22 (1): 80–85. <https://doi.org/10.1093/eurpub/ckr051>.
- Regidor, Enrique, Laura Reques, María José Belza, et al. 2014. “Education and Mortality in Spain: A Review of the Literature.” *Gaceta Sanitaria* 28 (6): 492–98. <https://doi.org/10.1016/j.gaceta.2014.06.001>.